1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Among localized prostate cancers with cribriform gland architecture, tumors exhibiting high AR signaling activity (as indicated by elevated AR expression and a robust AR-target transcriptional program) will be more sensitive to androgen-deprivation therapy (ADT), showing greater PSA decline at defined early timepoints and longer biochemical progression-free survival compared with cribriform-pattern tumors with low AR signaling activity. A composite ME-ARS score that integrates a normalized Cribriform Index (CI) with AR activity (AR expression plus AR-target transcriptional score) will predict ADT response more accurately than CI or AR metrics alone. Proliferation modifiers (e.g., MKI67, PCNA) will modulate this relationship, with high proliferation potentially reducing AR-dependence and attenuating ADT response within cribriform tumors. This framework supports stratification for neoadjuvant or definitive ADT in localized disease and clarifies when cribriform morphology indicates true AR-driven vulnerability versus AR-independent resistance.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: Assemble a cohort of localized prostate cancer patients treated with ADT (with available pretreatment tissue) and collect outcome data on PSA response and progression-free metrics.

Prediction 2: Quantify CI from digital pathology of pretreatment slides; quantify AR activity via (a) AR protein expression by IHC and (b) an AR-target gene expression score from RNA-seq or targeted panels; compute a composite AR activity score.

Prediction 3: Build a ME-ARS model testing the interaction between CI and AR activity to predict therapy_response; include MKI67 and PCNA as secondary modifiers to explore proliferation-driven resistance.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions